

Speaker Identification — Pilot Validation Results

Smart Subtitle Glasses FYP · Sprint 1 Speaker Embedding Experiment

Ayesha & Sabeeh

September 2026

100%

Known-speaker accuracy
after threshold calibration

100%

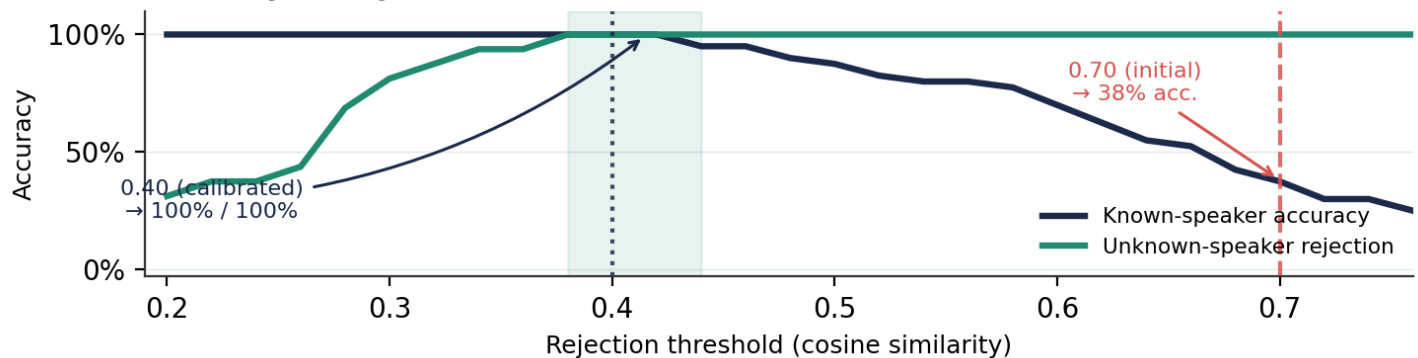
Stranger rejection
0 false accepts, all durations

0.70 → 0.40

Rejection threshold
recalibrated on held-out data

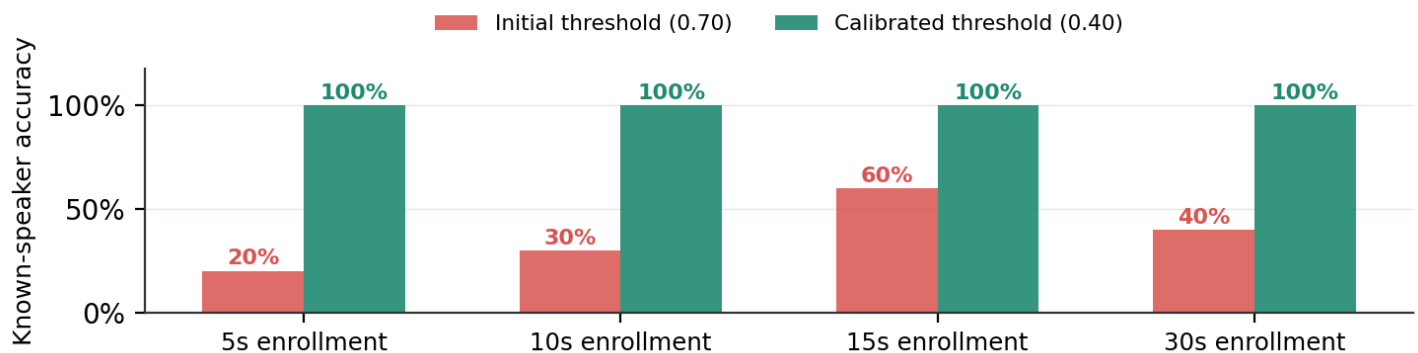
Finding 1 — A single recalibration fixes the accuracy problem

Accuracy vs. rejection threshold



Finding 2 — Enrollment length barely matters for correctness

Accuracy by enrollment length — before vs. after calibration



Key Findings

- The embedding model ranked the **correct enrolled speaker as the closest match in 100% of comparisons** (56/56), at every enrollment length tested (5s–30s) — the two voices are reliably distinguishable.
- The original low accuracy (20–60%) at threshold 0.70 was a **calibration issue, not a discrimination failure**: Sabeeh's genuine-speaker similarity scores typically fell between 0.43–0.71, under the 0.70 cutoff, so correct matches were being rejected as “unknown.”
- Recalibrating the threshold to **0.40** (chosen empirically from this test set) restores **100% identification accuracy** for both speakers at every enrollment length, while keeping stranger rejection at a perfect **100%** (0 of 7 unknown clips falsely accepted).
- Enrollment duration did not affect ranking correctness — even **5-second enrollments** correctly identified their speaker every time. Longer enrollment increases confidence margin, not correctness.

Scope of this pilot: 2 enrolled speakers, 10 known-speaker test clips, 7 unknown-speaker clips, clean audio. The 0.40 threshold should be re-validated as more speakers, background noise, distance, and Urdu-language samples are added in the next round of testing (per Sprint 1 plan).